

# Assignment 2-Case Study

## Stage 3 Tutorial Activities

### From Requirements to Domain Models

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Week 6 | 60 minutes

#### Candidate Concepts

| Candidate    | Class? | Reason                                                |
|--------------|--------|-------------------------------------------------------|
| Patient      | Yes    | SmartCare requires managing information about patient |
| Practitioner | Yes    | Practitioners need to be managed by the system        |
| Appointment  | Yes    | Main part of the system                               |
| Name         | No     | It is an attribute                                    |
| Clinic       | No     | Not in the requirements                               |
| Database     | No     | It's a technical issue                                |
| Cancellation | No     | Separate cancellation is not required                 |
| Status       | No     | It is not required                                    |

#### CRC Cards

##### Patient

|                                     |               |
|-------------------------------------|---------------|
| Responsibilities                    | Collaborators |
| Storing patient data or information | Appointment   |
| Associated with appointment         | Appointment   |

##### Practitioner

|                                            |               |
|--------------------------------------------|---------------|
| Responsibilities                           | Collaborators |
| Storing practitioners' data or information | Appointment   |
| Associated with appointment                | Appointment   |

##### Appointment

|                                       |               |
|---------------------------------------|---------------|
| Responsibilities                      | Collaborators |
| Storing appointment time              | Patient       |
| Connect practitioner with the patient | Practitioner  |

#### Relationship Reasoning

Patient to Appointment: which relationship and why?

- Association because a patient can have appointments and an appointment is linked to a patient.

Practitioner to Appointment: what multiplicity?

- 1-0 because a practitioner can zero or many appointments while an appointment is linked to one practitioner.

Should Appointment inherit from Patient?

- No, appointment is not a type of patient, its just associated with patient.

Does Clinic need to own every object?

- As per current mentioned object, patient, practitioners or appointment, clinic does not need to own every object.

## AI Model Critique

Critique AI proposals: PatientManager, PractitionerManager, AppointmentManager, ClinicController, NotificationManager, ScheduleEngine.

| AI proposal         | Decision | Reason                                                                                             |
|---------------------|----------|----------------------------------------------------------------------------------------------------|
| PatientManager      | ✗ Reject | The requirements identify <b>Patient</b> , but do not require a separate manager.                  |
| PractitionerManager | ✗ Reject | <b>Practitioner</b> is already a required concept; a manager is not supported by the requirements. |
| AppointmentManager  | ✗ Reject | <b>Appointment</b> is required, but there is no requirement for a separate manager class.          |
| ClinicController    | ✗ Reject | The requirements do not state that a controller class is needed.                                   |
| NotificationManager | ✗ Reject | No confirmed requirement for a notification system.                                                |
| ScheduleEngine      | ✗ Reject | No confirmed requirement for an automated scheduling engine.                                       |